Does Cross-Attention Fusion of ECG and Pulse Signals Verify ICU Arrhythmia Alarms? A Leakage-Audited Study on PhysioNet/CinC 2015 and VTaC

[Author 1], [Author 2], [Author 3]
[Department, Institution]
City, Country
ail addresses

***Abstract*—Intensive-care monitors raise many false arrhythmia alarms, and the resulting alarm fatigue can delay the response to true events. This study asks whether an independent pulse signal (photoplethysmogram, PPG) can verify an ECG-triggered alarm, and whether cross-attention fusion of the two signals is the right way to do it. We compared 14 neural architectures, six classical algorithms on 266 engineered features, a hybrid attention CNN–GRU recipe and ensembles on two open datasets, PhysioNet/CinC Challenge 2015 (592-record PPG cohort) and VTaC (4,542 ventricular tachycardia alarms), using a leakage-free protocol with inner validation sets, 10 seeds and corrected statistics. Four findings emerge. First, removing a validation leak from our initial code lowered the F1 of the proposed cross-attention model from 0.774 to 0.699 and erased its advantage over concatenation (ΔF1 $= +0.006$, 95% CI -0.015 to $+0.028$). Second, engineered-feature classifiers (F1 up to 0.818) and the hybrid deep recipe (F1 0.828 on Challenge 2015; F1 0.858 and AUC 0.966 on VTaC) clearly outperform plain cross-attention. Third, counterfactual tests show that fused models ignore ECG-to-pulse timing: substituting another patient's same-label PPG changes AUC by less than 0.03, and attention does not track measured pulse transit time ($\rho = -0.04$). Fourth, transfer across datasets is asymmetric: models trained on Challenge 2015 reach F1 of only 0.56–0.60 on VTaC when the ECG is used. We conclude that the pulse acts as a patient-agnostic evidence cue rather than a timing check, that training and feature engineering matter more than the fusion layer, and that single-dataset results overstate generalization. This is a research prototype without prospective validation.**

***Index Terms*—false alarm reduction, intensive care unit, arrhythmia alarm verification, multimodal fusion, cross-attention, electrocardiogram, photoplethysmogram, data leakage, pulse transit time, PhysioNet**

I. Introduction

Bedside monitors in intensive care units (ICUs) generate a large volume of arrhythmia alarms, and a considerable share of them are false. Observational work on consecutive ICU patients and reviews of the alarm-fatigue literature report that this burden desensitizes staff and can slow the response to true alarms [1], [2]. Ventricular tachycardia (VT) alarms are among the most frequent life-threatening alarms, and false VT alarms have proved the hardest to suppress reliably [3], [4]. Most monitors raise an alarm from the electrocardiogram (ECG) alone. Artifacts that corrupt the ECG, such as electrode motion or muscle noise, therefore also corrupt the signal that would be used to check the alarm.

A second, physically independent signal offers a way out. The photoplethysmogram (PPG) and the arterial blood pressure (ABP) waveform report whether an electrical event was followed by a mechanical pulse, and they fail under different conditions than the ECG. A genuine arrhythmia should disturb both signals, whereas an ECG artifact should leave the pulse intact. Earlier work exploited this idea with hand-crafted ABP features [5], in the 2015 PhysioNet/Computing in Cardiology Challenge [4], [6], and with deep models. The most closely related deep models are the attention-based convolutional and recurrent networks of Mousavi et al. [7], which reported a sensitivity of 93.88% and a specificity of 92.05% with three signals, and the contrastive approach of Zhou et al. [8], which adds learned alarm-type embeddings.

Our project started from a specific hypothesis: that a transformer-style *cross-attention* layer [9], in which ECG features query PPG features, can learn the pulse transit time (PTT) between the electrical beat and the arriving pulse, and that this comparison verifies alarms better than a model that sees only one signal or simply concatenates both. The research problem is therefore twofold: (i) does fusing ECG and pulse improve alarm verification, and (ii) if it does, is the gain explained by the attention-based alignment that motivated the design? A first version of our code answered (ii) positively (F1 of 0.774 for cross-attention), but that estimate was optimistically biased because the best epoch was selected on the same fold that was reported. Re-running the study under a leakage-free protocol is the starting point of this paper.

The objectives are to: (1) compare cross-attention with ECG-only, PPG-only and concatenation baselines at matched parameter count; (2) benchmark it against stronger classical, deep, hybrid and ensemble alternatives on two datasets; (3) test with counterfactual perturbations whether fusion models use ECG-to-pulse timing; (4) measure robustness to noise and sensor loss and transfer across datasets; and (5) report all results with corrected statistics, including those that contradict

the original hypothesis.

The remainder of the paper is organized as follows. Section II describes the datasets, preprocessing, models, training procedure, implementation and mathematical formulation. Section III presents the results and discusses them, including limitations. Section IV concludes.

II. Methods

A. Overview

Fig. 1 summarizes the pipeline. Each alarm event is represented by a 10 s window of ECG and a pulse channel that ends exactly at the alarm onset. Three families of models are trained on the same windows and splits: neural networks on the raw signals, classical learners on engineered features, and a hybrid deep recipe; their predictions are combined in ensembles.

B. Data collection

No new recordings were collected; all data are public, de-identified and were obtained from their original hosts. Table I lists every dataset with its citation and source URL. Cohort statistics below come from the audit files written by our preprocessing scripts.

Challenge 2015. The training set contains 750 five-minute ICU records at 250 Hz; the alarm is raised at the 300 s mark, and every alarm carries an expert true or false label for one of five types (asystole, extreme bradycardia, extreme tachycardia, ventricular tachycardia, ventricular flutter/fibrillation) [4], [6]. The 500-record test set was never released, so record-wise cross-validation is the appropriate evaluation. Of the 750 records, 627 carry PPG, 343 carry ABP and all have a second ECG lead. The original *PPG cohort* has 592 records (37.7% true alarms). A *recovered cohort* of 724 records (39.6% true) substitutes ABP for the pulse where PPG is absent (132 records) and excludes 26 records whose window is too short or nearly flat-lined.

VTaC. VTaC contains 5,037 annotated VT alarm events with at least two independent expert annotators each [3]; v1.1 additionally releases per-annotator votes. The host throttles each connection, so we downloaded only the window $[-60\,\mathrm{s}, +30\,\mathrm{s}]$ around each alarm onset using parallel HTTP range requests (about 1 GB instead of 18 GB). After excluding 292 events without a pulse channel and 203 low-quality windows, 4,542 events from 2,138 patients remain (28.0% true alarms); the official patient-disjoint split yields 3,662 training, 453 validation and 427 test events. Hospital and device identifiers are not released.

Auxiliary data. MIMIC PERform AF (35 subjects, 125 Hz) is used only for an AF-screening task, with 60 evenly spaced 10 s windows per subject (2,100 windows) [12], [13]. The MIT-BIH Noise Stress Test supplies real noise for stress testing [14].

C. Data preparation

Windowing. The headline task window is the 10 s that end exactly at the alarm onset; no post-alarm samples enter any model input. The 30 s after the onset are read only by a separate time-to-verdict analysis. This decision-time alignment replaced an earlier center-aligned window.

Preprocessing. Signals are resampled to 250 Hz and filtered with a zero-phase fourth-order Butterworth band-pass (ECG 0.5–40 Hz; pulse 0.5–8 Hz), then z-normalized per window, so that nothing is fitted across windows and no train/test statistics leak. A simple per-channel quality score (flat-line variance and clipping fraction) removes windows below a quality threshold of 0.05.

Splits. Challenge 2015 uses record-wise 5-fold cross-validation repeated for 10 seeds (the seed changes the fold assignment). VTaC uses the official patient-disjoint split with 5 to 10 seeds. Inside every training fold, 15% of the records form an *inner validation set* that drives early stopping, model selection, calibration, thresholds and ensemble weights; the outer test fold is evaluated once. All models see identical splits, so comparisons are paired.

D. Model selection

Parameter-matched baseline ladder. The core comparison uses four variants matched to about 261k parameters (single-signal models are widened): ECG-only, PPG-only, concatenation fusion, and the proposed cross-attention fusion. Each signal passes through its own (not weight-shared) 1D-CNN encoder (channels 32/64/128, kernel sizes 15/9/7, pooling 2, dropout 0.3) that outputs a feature *sequence*. In cross-attention (4 heads, dimension 128, positional encodings) the ECG sequence supplies the queries and the PPG sequence the keys and values; a shallow head (64 hidden units) produces the alarm logit.

Rivals. We added four fusion variants (bidirectional cross-attention, joint transformer with modality embeddings, bottleneck-token fusion, quality-gated cross-attention) and a baseline zoo on a 2-channel early-fusion input: ResNet1D [17], InceptionTime [18], a bidirectional GRU, a TCN [19], a patch transformer and a compact diagonal state-space model. In total 14 neural architectures were compared.

Classical models. 266 engineered features (HRV, ECG morphology proxies, wavelet and spectral features, PPG pulse features, ECG–PPG agreement such as PTT statistics and beat-count ratios in the last 5 s, and signal-quality indices) feed six algorithms: logistic regression, RBF-SVM, random forest, extra trees, XGBoost [20] and an MLP. ECG beats are detected with a Pan–Tompkins-style detector [21]. Thirty random hyper-parameter configurations per fold and algorithm are scored on the inner validation set only. A first, 53-feature gradient-boosting baseline ("features v1") is also reported.

Hybrid deep recipe. Following the attention CNN–RNN design of [7] (our re-implementation from the published description, not the authors' code), a 48-configuration grid (6 architectures $\times$ augmentation $\times$ loss $\times$ learning rate) was declared in advance and scored by inner-validation AUC. The winner, an attention CNN–GRU with augmentation, focal loss and learning rate 10^{-3}, was combined with AdamW [22], a one-cycle schedule [23], weight EMA, test-time augmentation

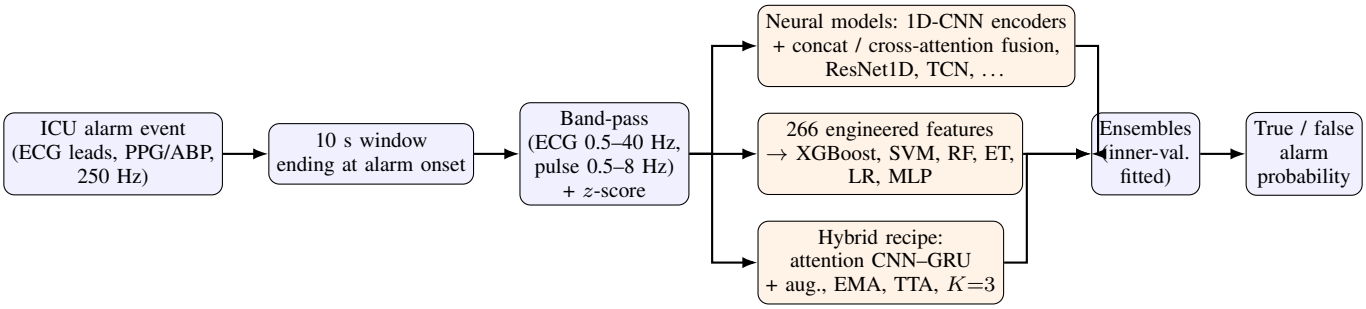


Fig. 1. Overview of the verification pipeline. All stages see only samples up to the alarm onset; hyper-parameters, thresholds and ensemble weights are fitted on inner validation records of each training fold.

TABLE I
DATASETS USED IN THIS STUDY, WITH REQUIRED CITATIONS AND SOURCE URLS.

Dataset	Role	Content used	Citation and source URL
PhysioNet/CinC Challenge 2015 training set	Primary training and cross-validation	750 ICU records, 5 alarm types, expert true/false label; 592-record PPG cohort, 724-record recovered cohort	[6], [10]; https://physionet.org/content/challenge-2015/1.0.0/
VTaC v1.1	Second training/test set, official patient-disjoint split	5,037 annotated VT alarms (3 US hospitals, 3 monitor vendors); 4,542 used	[3], [11]; https://physionet.org/content/vtac/
MIMIC PERform AF	External AF-screening task (35 subjects)	19 AF, 16 non-AF subjects; ECG + PPG at 125 Hz	[12], [13]; https://zenodo.org/records/15906524
MIT-BIH Noise Stress Test	Synthetic noise injection (robustness)	Baseline wander, electrode motion, muscle artifact	[14]; https://physionet.org/content/nstdb/1.0.0/
MIT-BIH Arrhythmia Database	Validation of the R-peak detector	10 records	[15]; https://physionet.org/content/mitdb/1.0.0/
PaPaGei pretrained weights	Foundation-model comparison (not a dataset)	PPG encoders PaPaGei-S/-P	[16]; https://zenodo.org/records/13983110

and a 3-member ensemble. Hybrid variants that append the 53 agreement features to the deep representation were also tested.

Ensembles. Logit mean, probability mean, a logistic-regression stacker and greedy selection are all fitted on inner-validation predictions only.

Self-supervised pretraining and foundation models. Contrastive, masked-reconstruction and cross-modal-prediction objectives were pretrained on VTaC pre-alarm segments of training/validation patients only. The PaPaGei PPG foundation model [16] was evaluated frozen (linear probe), as a frozen encoder inside fusion models, and with last-block fine-tuning. ECG-FM [24] weights were downloaded but not run, because the model needs a 12-lead 500 Hz input and a compiler toolchain that was unavailable.

E. Training procedure

Baseline-ladder models use Adam with learning rate 10^{-3}, weight decay 10^{-4}, batch size 16, at most 60 epochs and early stopping (patience 10) on inner-validation F1, with focal loss. The hybrid recipe uses AdamW (weight decay 10^{-2}), a one-cycle schedule with 15% warm-up, gradient clipping at 1.0, batch 32 (64 on VTaC), at most 30 epochs (20 on VTaC) and early stopping on inner-validation AUC. Augmentation (training only) comprises a joint circular time shift of ± 1 s, per-channel amplitude scaling, additive Gaussian noise and a random 1 s dropout of one modality. Hyper-parameters of the ladder are fixed in a configuration file and not tuned per variant. Class imbalance is handled by class weighting or focal loss, not resampling.

F. Implementation details and tools

Code is written in Python with PyTorch [25], scikit-learn [26], XGBoost [20], SciPy, PyWavelets, pandas and the WFDB package for reading PhysioNet records [27]. Experiments ran on a single workstation GPU (RTX PRO 4000 Blackwell, 24 GB) with 20 CPU cores; neural folds train in seconds to minutes. The repository holds the code, configurations and the result files from which every table and figure here was produced.

G. Mathematical formulation

Let $E, P \in \mathbb{R}^{T' \times d}$ be the ECG and PPG feature sequences (with positional encodings added). Cross-attention computes

$$\mathrm{Attn}(E, P) = \mathrm{softmax}\left(\frac{(EW_Q)(PW_K)^\top}{\sqrt{d_k}}\right) PW_V, \quad (1)$$

so each ECG time step aggregates the PPG steps judged relevant. Concatenation fusion instead stacks the pooled features. Training uses the focal loss [28]

$$\mathcal{L} = -\alpha_t (1 - p_t)^{\gamma} \log p_t, \quad \gamma = 2, \tag{2}$$

where p_t is the predicted probability of the true class. The pulse transit time of beat i is $\mathrm{PTT}_i = t_i^{\mathrm{foot}} - t_i^{R}$ (R-peak to PPG foot). Classification is scored by $\mathrm{F1} = 2\,\mathrm{TP}/(2\,\mathrm{TP} + \mathrm{FP} + \mathrm{FN})$, AUC, and the Challenge 2015 score

$$S = \frac{\mathrm{TP} + \mathrm{TN}}{\mathrm{TP} + \mathrm{TN} + \mathrm{FP} + 5\,\mathrm{FN}}, \tag{3}$$

in which a missed true alarm costs five false alarms [6]. For two models evaluated on J paired folds, the Nadeau–Bengio corrected resampled t-statistic [29] is

$$t = \frac{\bar{d}}{\sqrt{\left(\frac{1}{J} + \frac{n_{\mathrm{test}}}{n_{\mathrm{train}}}\right) s_d^2}}, \tag{4}$$

with $\bar{d}$ and s_d^2 the mean and variance of the paired F1 differences. We additionally use a paired record-level bootstrap with 5,000 resamples [30], in which the same resample is applied to both models, and Holm correction [31] over a comparison family fixed in the configuration file before the final runs. A finding is called robust only if both tests agree after correction. The safety layer applies temperature scaling [32] and certifies a suppression threshold with an exact Clopper–Pearson upper bound on the miss rate [33]. The repository also contains a TRIPOD+AI reporting checklist [34].

III. Results and Discussion

A. Effect of removing the validation leak

The original training script selected the best epoch and stopped training on the same fold it then reported. With a separate inner validation set and 10 seeds, every score fell (Fig. 2): the cross-attention F1 dropped from 0.774 to 0.699 and its advantage over concatenation shrank from $+0.024$ to $+0.006$ (bootstrap 95% CI -0.015 to $+0.028$). Because the leak is a property of our earlier code and not of the method, we archived the original numbers and do not use them elsewhere.

B. Leaderboards

Tables II and III and Fig. 3 give the main comparison (threshold 0.5; means over seeds and folds).

On Challenge 2015 the proposed cross-attention network ranks 12th of the 13 models in Table II by F1, behind every classical model on engineered features and behind ResNet1D, TCN and InceptionTime. The best leak-free results come from the 10-model ensemble (accuracy 0.876) and from the hybrid deep recipe (F1 0.828, AUC 0.932). The ensemble's F1 advantage over the best classical model is within noise ($+0.008$, CI -0.007 to $+0.022$), but its advantage over cross-attention is $+0.128$ (CI $+0.094$ to $+0.161$). On VTaC the hybrid recipe leads (F1 0.858, AUC 0.966); differences among the top three models are inside bootstrap noise, whereas the gap to cross-attention is $+0.11$ F1 (CI $+0.07$ to $+0.15$).

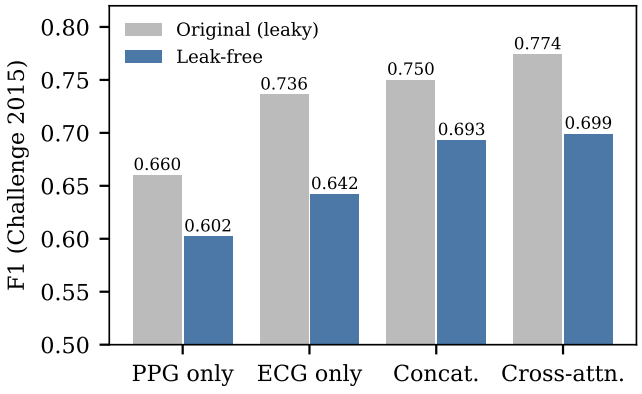


Fig. 2. F1 on Challenge 2015 before and after removing the validation leak.

TABLE II
Challenge 2015, PPG cohort (592 records), 5-fold CV × 10 seeds.

Model	Acc.	F1	AUC
Ensemble, 10 models (logit mean)	0.876	0.827	0.948
XGBoost, features v2	0.871	0.818	0.939
Extra Trees, features v2	0.867	0.802	0.947
Random Forest, features v2	0.863	0.795	0.938
MLP, features v2	0.857	0.802	0.927
SVM (RBF), features v2	0.852	0.791	0.928
Logistic reg., features v2	0.852	0.801	0.928
HistGB, features v1	0.851	0.798	0.918
ResNet1D	0.820	0.760	0.884
TCN	0.810	0.751	0.887
InceptionTime	0.796	0.732	0.876
CNN cross-attention (proposed)	0.765	0.699	0.841
CNN concatenation	0.757	0.693	0.836
Hybrid deep recipe (10 seeds)	0.871	0.828	0.932
Ensemble incl. hybrid (5 seeds)	0.878	0.830	0.950

Fusion does help on VTaC: ECG-only reaches F1 0.709 and PPG-only 0.566, against 0.744 for concatenation.

Caveat on the hybrid recipe. Its configuration was selected on inner-validation scores of Challenge 2015 folds. Those records are test records in other seeds and folds, so a small optimistic bias is possible. The same recipe was applied to VTaC without re-tuning, and a shuffled-label control gave a chance-level AUC of 0.57, so we found no evidence of leakage. A recipe ablation (Table IV) shows that the largest single post-training ingredient is weight averaging (EMA), and part of the gap to the baseline CNNs is a trainer effect (optimizer, regularization, ensembling) rather than an architecture effect, because the baselines use a different trainer.

C. Pre-specified paired comparisons

Table V and Fig. 4 report the comparison family fixed before the final runs.

The four attention-style fusion variants are statistically indistinguishable from each other and from concatenation. The bootstrap supports a benefit of fusion over either single

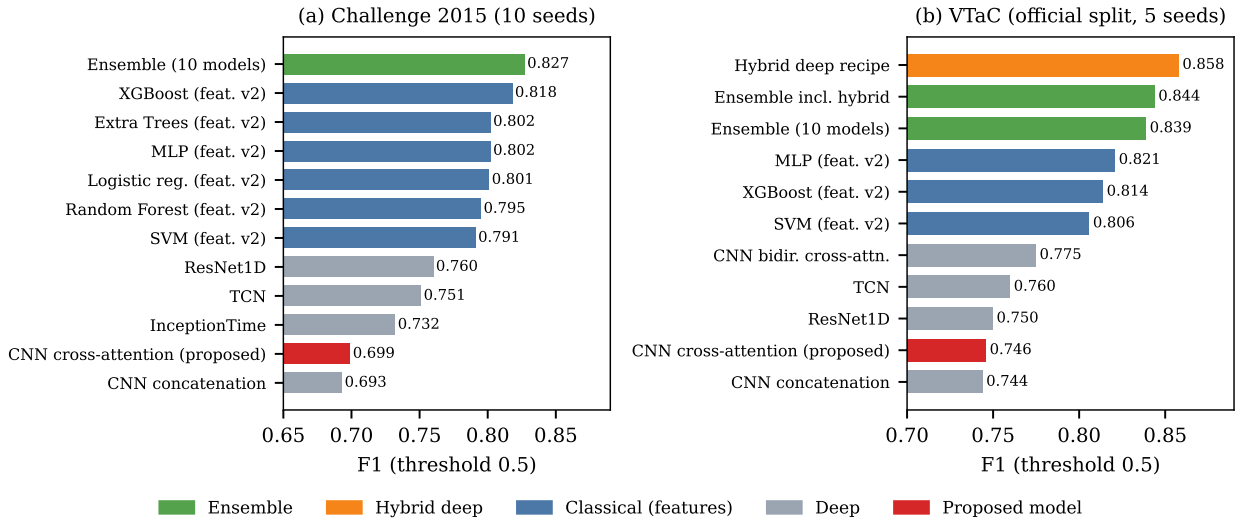


Fig. 3. F1 leaderboards (threshold 0.5). The originally proposed cross-attention model (red) ranks near the bottom on both datasets.

TABLE III
VTaC, OFFICIAL SPLIT (427 TEST EVENTS), 5 SEEDS.

Model	Acc.	F1	AUC
Hybrid deep recipe	0.913	0.858	0.966
Ensemble incl. hybrid (logit mean)	0.912	0.844	0.957
Ensemble, 10 models (logit mean)	0.910	0.839	0.955
MLP, features v2	0.900	0.821	0.949
XGBoost, features v2	0.895	0.814	0.942
SVM (RBF), features v2	0.891	0.806	0.948
CNN bidirectional cross-attention	0.857	0.775	0.933
TCN	0.856	0.760	0.919
ResNet1D	0.852	0.750	0.915
CNN cross-attention (proposed)	0.844	0.746	0.915
CNN concatenation	0.837	0.744	0.909

TABLE IV
ABLATION OF THE HYBRID RECIPE, CHALLENGE 2015 (10 SEEDS, PAIRED AGAINST THE FULL RECIPE; SE OVER SEEDS).

Variant	F1	AUC	ΔF1 (SE)
Full recipe (K=3, EMA, TTA)	0.828	0.932	0
No TTA	0.824	0.931	−0.003 (0.002)
No EMA	0.819	0.932	−0.008 (0.002)
Ensemble size 2	0.828	0.930	0.000 (0.002)
Single member (EMA, TTA)	0.815	0.925	−0.012 (0.004)
Single member, raw, no TTA	0.793	0.922	−0.035 (0.010)

signal, but the more conservative Nadeau–Bengio test does not after Holm correction, so by our own rule (both tests must agree) only comparisons such as GBM versus cross-attention are called robust. This asymmetry between the two tests is expected, since the corrected t-test accounts for overlapping training sets.

TABLE V
PAIRED F1 DIFFERENCES ON CHALLENGE 2015 (CHALLENGER − BASELINE; 50 SEED×FOLD PAIRS). p_{NB}: NADEAU–BENGIO, p_{boot}: BOOTSTRAP; BOTH HOLM-ADJUSTED.

Baseline	Challenger	ΔF1	95% CI	p_{NB}	p_{boot}
Concat.	Cross-attn.	+0.006	[−0.015, +0.028]	1.00	1.00
ECG-only	Cross-attn.	+0.058	[+0.028, +0.088]	1.00	0.003
PPG-only	Cross-attn.	+0.094	[+0.058, +0.129]	0.077	<0.001
Cross-attn.	Bidirectional	−0.008	[−0.023, +0.008]	1.00	1.00
Cross-attn.	Joint transf.	−0.006	[−0.025, +0.013]	1.00	1.00
Cross-attn.	Gated	−0.002	[−0.015, +0.012]	1.00	1.00
Cross-attn.	Bottleneck	−0.047	[−0.070, −0.026]	1.00	<0.001
ECG-only	ResNet1D	+0.118	[+0.089, +0.148]	0.052	<0.001
ResNet1D	Cross-attn.	−0.060	[−0.087, −0.035]	0.95	<0.001
TCN	Cross-attn.	−0.052	[−0.080, −0.023]	1.00	<0.001
GBM (feat. v1)	Cross-attn.	−0.099	[−0.133, −0.066]	0.049	<0.001

D. *Per-alarm-type behavior*

Table VI shows pooled out-of-fold results for cross-attention. Performance is highly type-dependent: tachycardia has an F1 of 0.92 but a specificity of only 0.42, VT has F1 0.49, and asystole and ventricular flutter/fibrillation have too few true alarms (14 and 5) for reliable conclusions. In the error taxonomy, persistent missed true alarms are dominated by VT (18 of 29 persistent false negatives for the feature GBM, although VT is only 26% of true alarms).

E. *Do fusion models use ECG-to-pulse timing?*

This is the question behind the original design. Perturbing one modality while the other is intact (seed-0 checkpoints, five test folds; Fig. 5) shows that replacing the PPG by another patient's *same-label* PPG, whose timing is unrelated to the ECG, changes AUC by only 0.00–0.03 (cross-attention 0.002, concatenation 0.015, ResNet1D 0.021, TCN 0.024). Shifting the PPG by up to 2 s changes AUC by at most 0.005 for the CNN fusion models and by 0.03–0.06 for ResNet1D, TCN,

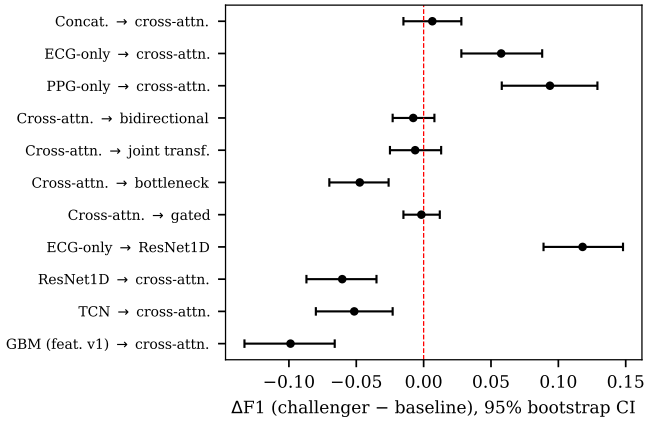


Fig. 4. Paired F1 differences with 95% bootstrap confidence intervals (Challenge 2015).

TABLE VI
CROSS-ATTENTION PER ALARM TYPE (CHALLENGE 2015, POOLED OUT-OF-FOLD).

Alarm type	n	True	F1	Sens.	Spec.
Asystole	93	14	0.41	0.56	0.80
Bradycardia	73	39	0.75	0.69	0.83
Tachycardia	111	106	0.92	0.87	0.42
V. flutter/fib.	45	5	0.26	0.42	0.77
V. tachycardia	270	59	0.49	0.58	0.78

InceptionTime and BiGRU. In contrast, an *opposite-label* pulse collapses AUC (cross-attention 0.842 to 0.535). The models therefore use the *content* of each signal (morphology, regularity, presence of a pulse) as evidence, not the beat-to-pulse timing.

The attention maps agree. The attention-implied lag shows no relationship with the measured PTT (window-level Spearman $\rho = -0.039$, permutation $p = 0.39$; beat-level $\rho = 0.017$), and only 5.3% of attention mass falls in the physiological lag range versus 5.6% for uniform attention (Fig. 6). In deletion/insertion tests, occlusion is clearly faithful (PPG deletion -0.041, insertion $+0.041$) and integrated gradients modestly so (-0.016/$+0.010$), whereas attention weights are barely better than random ordering (-0.007/$+0.011$). Attention weights should therefore not be presented as an explanation, and the pulse-transit-time justification in our original proposal is not supported by these data.

F. Robustness to noise and sensor loss

With real Noise Stress Test noise on the ECG and synthetic motion artifact on the PPG (SNR 24 to 0 dB), ResNet1D is the most robust (AUC 0.907 clean to 0.844 at 0 dB on both signals), whereas cross-attention falls from 0.856 to 0.696 and concatenation from 0.833 to 0.720 (Fig. 7). Zeroing the PPG at inference, without retraining, costs cross-attention 0.12 AUC (0.856 to 0.736), concatenation 0.10 and ResNet1D 0.27; zeroing the ECG costs cross-attention 0.08. Constant PPG

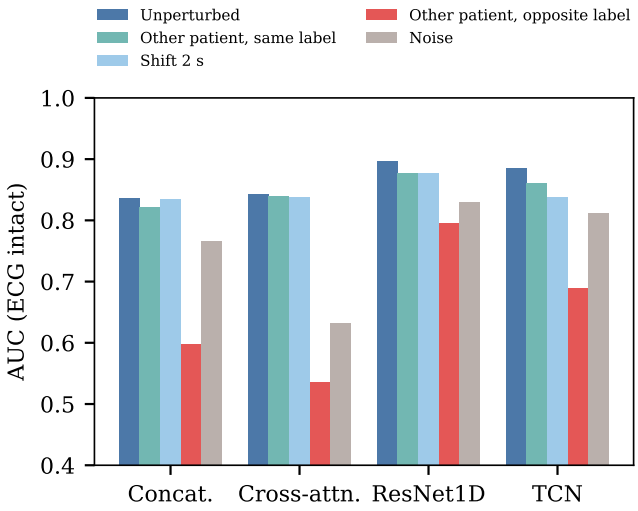


Fig. 5. Test AUC when the PPG is perturbed and the ECG is intact.

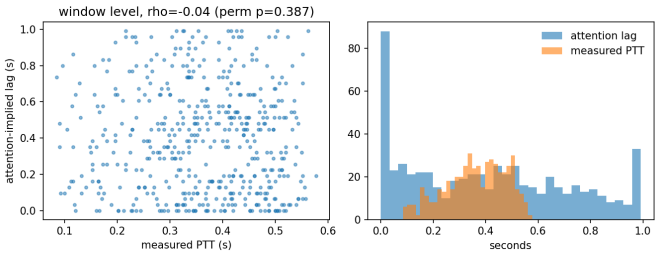


Fig. 6. Attention-implied lag versus measured PTT (left, window level) and the two lag distributions (right).

delays of 0–400 ms barely change the CNN fusion models, consistent with the previous subsection. A check that the injected noise is not a learnable shortcut gave an AUC of 0.52–0.55 from model probabilities alone.

G. Generalization across datasets

Transfer is strongly asymmetric (Fig. 8). Models trained on the small Challenge 2015 set and tested on the VTaC test split reach F1 of 0.56–0.60 (AUC 0.73–0.78) when the ECG is used, whereas the PPG-only model is near chance (F1 0.432, AUC 0.541); in-distribution VTaC F1 is 0.71–0.78. In the other direction, models trained on VTaC transfer well to Challenge 2015 VT alarms (ResNet1D F1 0.773, AUC 0.927). For comparison, cross-attention trained and tested within Challenge 2015 reaches a VT F1 of only 0.49, so the amount of training data matters more than the model family, and single-dataset numbers overstate real-world performance.

H. Pretraining, foundation models and AF screening

Fine-tuning from self-supervised encoders gives modest paired gains over scratch training (+0.01 to +0.04 F1, three seeds, label fractions 10–100%), but only 1 of 72 paired contrasts has a Nadeau–Bengio $p < 0.05$, so the effect is suggestive rather than established. The PaPaGei foundation

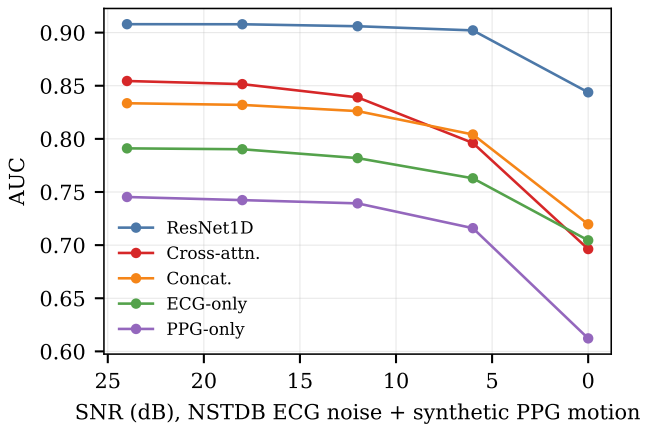


Fig. 7. AUC versus injected noise level (mean over noise types; seed-0 checkpoints).

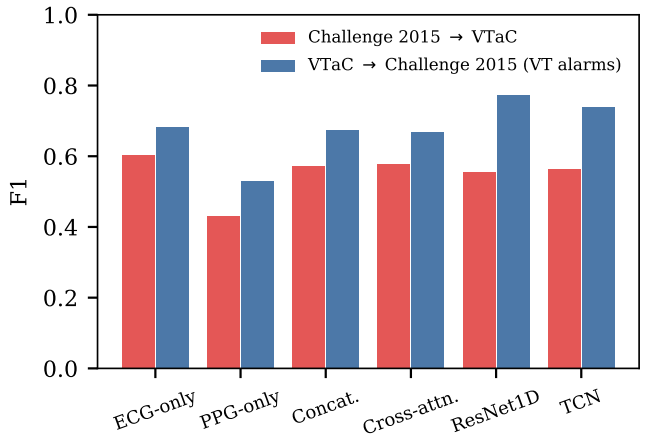


Fig. 8. Cross-dataset transfer (F1).

model did not beat from-scratch encoders: a frozen linear probe reached F1 0.58 (PPG alone) on Challenge 2015, and the best PaPaGei fusion model was about 0.04 F1 below scratch cross-attention; on VTaC it improved on a scratch PPG-only model (F1 0.58–0.62 versus 0.566) but not on scratch fusion (0.745). On the MIMIC PERform AF task (35 subjects, subject-wise cross-validation, low power) a gradient-boosting model on RR-irregularity features reached subject-level AUC 1.00 and was significantly better than the best neural model (0.93), while fusion beat a PPG-only network by $+0.28$ subject AUC. Distilling a PPG-only student from the fusion teacher gave no benefit over hard labels.

I. Label reliability, time-to-verdict and safety

On VTaC, 7.7% of events (351 of 4,542) have split annotator votes, and accuracy is much lower on them (cross-attention 0.704 versus 0.854 on unanimous events); model entropy tracks disagreement only weakly (Spearman 0.13–0.18). Training on annotator vote fractions improved cross-attention by $+0.038$ F1 (SD 0.012, 5 seeds). A time-to-verdict analysis slides the 10 s window past the trigger and shows F1 falling (cross-attention 0.740, 0.713, 0.515 and 0.451 at 0, 5, 10 and 30 s); because the triggering event leaves the window, this is not evidence about whether waiting helps, and a design that appends post-alarm seconds to the pre-alarm window was not run.

A safety layer (temperature scaling, a risk-controlled threshold with a Clopper–Pearson bound, and a deferral band) was run for the concatenation model only. A 10% miss tolerance at 90% confidence could be certified in 42 of 50 folds, suppressing 22.5% of false alarms while missing 2.9% of true alarms, and calibration reduced the expected calibration error from 0.167 to 0.121. Thresholds aimed at 95–100% validation sensitivity realized only about 92–97% on test folds, so sensitivity targets must not be assumed to transfer.

J. Limitations

(i) The data are small: fold-to-fold F1 standard deviation is about 0.05, so many differences between top models are inside noise, and the standard deviations reported over seeds$\times$folds are not confidence intervals. (ii) Several engineered features, and on Challenge 2015 the alarm type itself, encode the alarm definitions; this is not learned physiology, and an ablation without the alarm type was not run. (iii) The hybrid recipe carries a small selection-bias risk. (iv) The Mousavi-style and contrastive reproductions are re-implementations from the published descriptions; only a two-seed sanity run of the former exists, so we make no claim about the original methods' performance. (v) There is no prospective validation, all data are from US ICUs, effects of demographics and skin tone on optical PPG were not assessed, VTaC device identifiers are not released (the lead-set signature used as a proxy is unverified), and patient overlap between corpora cannot be checked. (vi) Leave-one-device-out evaluation, long pre-alarm context, ECG-FM and pretraining on larger corpora (PulseDB, VitalDB, MIMIC-III) were not run because of time or data access. (vii) The model must not be used to suppress alarms unsupervised; the data are public and de-identified under the sources' IRB approvals.

IV. Conclusion

We tested whether cross-attention fusion of ECG and pulse signals verifies ICU arrhythmia alarms better than simpler alternatives. Under a leakage-free protocol the answer is no: the originally proposed model is not better than naive concatenation and is clearly beaten by engineered-feature classifiers, a carefully trained hybrid attention CNN–GRU and ensembles (accuracy 0.878–0.913, F1 0.83–0.86). The two signals are nonetheless complementary, since fusion beats either signal alone on VTaC, but the models exploit the content of each signal rather than the beat-to-pulse timing that motivated the design. Counterfactual tests, attention–PTT analysis and noise stress tests all point in this direction. The most important methodological lesson is that an unnoticed validation leak had inflated the headline F1 by 0.075, and that the controlled protocol is what makes the remaining conclusions trustworthy.

The most valuable next steps are leave-one-site-out evaluation, larger unlabeled pretraining corpora, and prospective validation. Code, configurations and result files accompany this paper in the project repository.

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References

[1] B. J. Drew, P. Harris, J. K. Zègre-Hemsey, T. Mammone, D. Schindler, R. Salas-Boni, Y. Bai, A. Tinoco, Q. Ding, and X. Hu, "Insights into the problem of alarm fatigue with physiologic monitor devices: a comprehensive observational study of consecutive intensive care unit patients," *PLOS ONE*, vol. 9, no. 10, p. e110274, 2014.

[2] M. Cvach, "Monitor alarm fatigue: an integrative review," *Biomedical Instrumentation & Technology*, vol. 46, no. 4, pp. 268–277, 2012.

[3] L. H. Lehman, B. Moody, H. Deep, F. Wu, H. Saeed, L. McCullum, D. Perry, T. Struja, Q. Li, G. Clifford, and R. G. Mark, "VTaC: A benchmark dataset of ventricular tachycardia alarms from ICU monitors," in *Advances in Neural Information Processing Systems (NeurIPS), Datasets and Benchmarks Track*, vol. 36, 2023. [Online]. Available: https://proceedings.neurips.cc/paper_files/paper/2023/hash/7a53bf4e02022aad32a4019d41b3b476-Abstract-Datasets_and_Benchmarks.html

[4] G. D. Clifford, I. Silva, B. Moody, Q. Li, D. Kella, A. Chahin, T. Kooistra, D. Perry, and R. G. Mark, "False alarm reduction in critical care," *Physiological Measurement*, vol. 37, no. 8, pp. E5–E23, 2016.

[5] A. Aboukhalil, L. Nielsen, M. Saeed, R. G. Mark, and G. D. Clifford, "Reducing false alarm rates for critical arrhythmias using the arterial blood pressure waveform," *Journal of Biomedical Informatics*, vol. 41, no. 3, pp. 442–451, 2008.

[6] G. D. Clifford, I. Silva, B. Moody, Q. Li, D. Kella, A. Shahin, T. Kooistra, D. Perry, and R. G. Mark, "The PhysioNet/Computing in Cardiology Challenge 2015: Reducing false arrhythmia alarms in the ICU," in *Computing in Cardiology*, vol. 42, 2015, pp. 273–276.

[7] S. Mousavi, A. Fotoohinasab, and F. Afghah, "Single-modal and multi-modal false arrhythmia alarm reduction using attention-based convolutional and recurrent neural networks," *PLOS ONE*, vol. 15, no. 1, p. e0226990, 2020.

[8] Y. Zhou, G. Zhao, J. Li, G. Sun, X. Qian, B. Moody, R. G. Mark, and L. H. Lehman, "A contrastive learning approach for ICU false arrhythmia alarm reduction," *Scientific Reports*, vol. 12, p. 4689, 2022.

[9] A. Vaswani, N. Shazeer, N. Parmar, J. Uszkoreit, L. Jones, A. N. Gomez, Ł. Kaiser, and I. Polosukhin, "Attention is all you need," in *Advances in Neural Information Processing Systems*, vol. 30, 2017.

[10] G. Clifford, I. Silva, B. Moody, and R. G. Mark, "Reducing false arrhythmia alarms in the ICU: The PhysioNet/Computing in Cardiology Challenge 2015 (version 1.0.0)," PhysioNet, 2015. [Online]. Available: https://physionet.org/content/challenge-2015/1.0.0/

[11] L. w. Lehman, B. Moody, L. McCullum, H. Saeed, H. Deep, D. Perry, T. Struja, Q. Li, G. Clifford, and R. Mark, "VTaC: A benchmark dataset of ventricular tachycardia alarms from ICU monitors," PhysioNet, version 1.1, 2024. [Online]. Available: https://physionet.org/content/vtac/

[12] P. H. Charlton, "MIMIC PERform datasets (version 2.0)," Zenodo, 2025. [Online]. Available: https://zenodo.org/records/15906524

[13] P. H. Charlton, K. Kotzen, E. Mejía-Mejía, P. J. Aston, K. Budidha, J. Mant, C. Pettit, J. A. Behar, and P. A. Kyriacou, "Detecting beats in the photoplethysmogram: benchmarking open-source algorithms," *Physiological Measurement*, vol. 43, no. 8, 2022.

[14] G. B. Moody, W. E. Muldrow, and R. G. Mark, "A noise stress test for arrhythmia detectors," *Computers in Cardiology*, vol. 11, pp. 381–384, 1984. [Online]. Available: https://physionet.org/content/nstdb/1.0.0/

[15] G. B. Moody and R. G. Mark, "The impact of the MIT-BIH arrhythmia database," *IEEE Engineering in Medicine and Biology Magazine*, vol. 20, no. 3, pp. 45–50, 2001. [Online]. Available: https://physionet.org/content/mitdb/1.0.0/

[16] A. Pillai, D. Spathis, F. Kawsar, and M. Malekzadeh, "PaPaGei: Open foundation models for optical physiological signals," in *Proc. International Conference on Learning Representations (ICLR)*, 2025, arXiv:2410.20542. [Online]. Available: https://arxiv.org/abs/2410.20542

[17] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Computer Vision and Pattern Recognition*, 2016, pp. 770–778.

[18] H. I. Fawaz *et al.*, "InceptionTime: Finding AlexNet for time series classification," *Data Mining and Knowledge Discovery*, vol. 34, no. 6, pp. 1936–1962, 2020.

[19] S. Bai, J. Z. Kolter, and V. Koltun, "An empirical evaluation of generic convolutional and recurrent networks for sequence modeling," arXiv:1803.01271, 2018.

[20] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proc. ACM SIGKDD Int. Conf. Knowledge Discovery and Data Mining*, 2016, pp. 785–794.

[21] J. Pan and W. J. Tompkins, "A real-time QRS detection algorithm," *IEEE Transactions on Biomedical Engineering*, vol. BME-32, no. 3, pp. 230–236, 1985.

[22] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," in *Proc. International Conference on Learning Representations*, 2019.

[23] L. N. Smith and N. Topin, "Super-convergence: very fast training of neural networks using large learning rates," in *Artificial Intelligence and Machine Learning for Multi-Domain Operations Applications, Proc. SPIE*, vol. 11006, 2019.

[24] K. McKeen *et al.*, "ECG-FM: An open electrocardiogram foundation model," arXiv:2408.05178, 2024. [Online]. Available: https://arxiv.org/abs/2408.05178

[25] A. Paszke *et al.*, "PyTorch: An imperative style, high-performance deep learning library," in *Advances in Neural Information Processing Systems*, vol. 32, 2019.

[26] F. Pedregosa *et al.*, "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[27] A. L. Goldberger, L. A. N. Amaral, L. Glass, J. M. Hausdorff, P. C. Ivanov, R. G. Mark, J. E. Mietus, G. B. Moody, C.-K. Peng, and H. E. Stanley, "PhysioBank, PhysioToolkit, and PhysioNet: components of a new research resource for complex physiologic signals," *Circulation*, vol. 101, no. 23, pp. e215–e220, 2000.

[28] T.-Y. Lin, P. Goyal, R. Girshick, K. He, and P. Dollár, "Focal loss for dense object detection," in *Proc. IEEE Int. Conf. Computer Vision*, 2017, pp. 2980–2988.

[29] C. Nadeau and Y. Bengio, "Inference for the generalization error," *Machine Learning*, vol. 52, no. 3, pp. 239–281, 2003.

[30] B. Efron and R. J. Tibshirani, *An Introduction to the Bootstrap*. Chapman & Hall/CRC, 1994.

[31] S. Holm, "A simple sequentially rejective multiple test procedure," *Scandinavian Journal of Statistics*, vol. 6, no. 2, pp. 65–70, 1979.

[32] C. Guo, G. Pleiss, Y. Sun, and K. Q. Weinberger, "On calibration of modern neural networks," in *Proc. International Conference on Machine Learning*, 2017, pp. 1321–1330.

[33] C. J. Clopper and E. S. Pearson, "The use of confidence or fiducial limits illustrated in the case of the binomial," *Biometrika*, vol. 26, no. 4, pp. 404–413, 1934.

[34] G. S. Collins *et al.*, "TRIPOD+AI statement: updated guidance for reporting clinical prediction models that use regression or machine learning methods," *BMJ*, vol. 385, p. e078378, 2024.